

***i*EMA: High-Fidelity, Low-Burden self-report acquisition framework for In-the-Wild Stress Detection**

July 16th, 2026

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◆ The reliability of EMA depends on whether the system asks at the right moment

- ❑ Two components of iEMA
 - ❑ Informative Moment(IM) detection : estimates the heart-rate response expected
 - ❑ Active Incremental Experience Sampling (AIES) : reduces unnecessary interruptions
- ❑ Three contributions of in-the-wild stress sensing
 - ❑ Temporally misaligned self-report acquisition : key source of label unreliability
 - ❑ Self-report acquisition framework : residual-based informative-moment detection + uncertainty-aware survey selection
 - ❑ In-the-wild-to-lab cross-domain evaluation methodology : Fair comparison of label-acquisition strategies

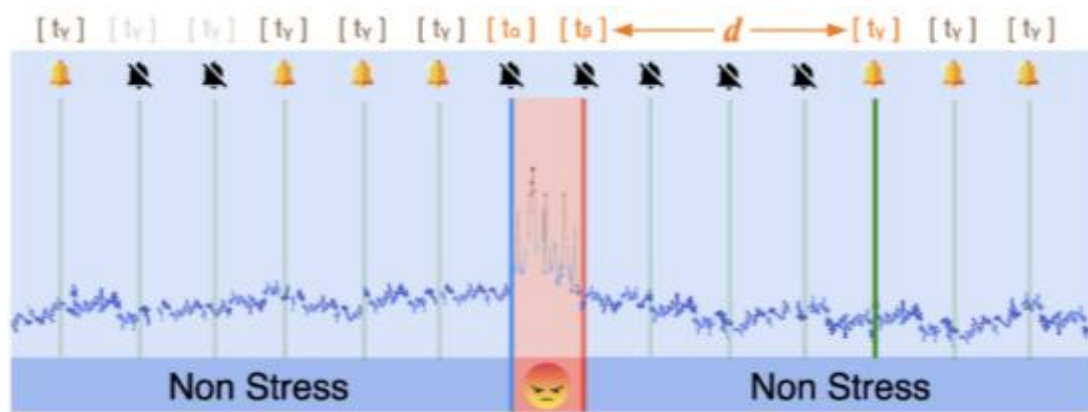
◆ Major bottleneck in real-world stress sensing

□ Relates to prior research in four areas

- 1) **EMA/ESM in the wild Settings** : focus on temporal fidelity
- 2) **Prompt strategy** : targets an event-centric prompting policy
- 3) **Adaptive Labeling** : Informative Moment(IM) module identifies candidate events
- 4) **In-the-Wild Stress Detection** : event-centric, burden-aware, and learning-oriented label acquisition

◆ Temporal Mismatch and Causal Ambiguity

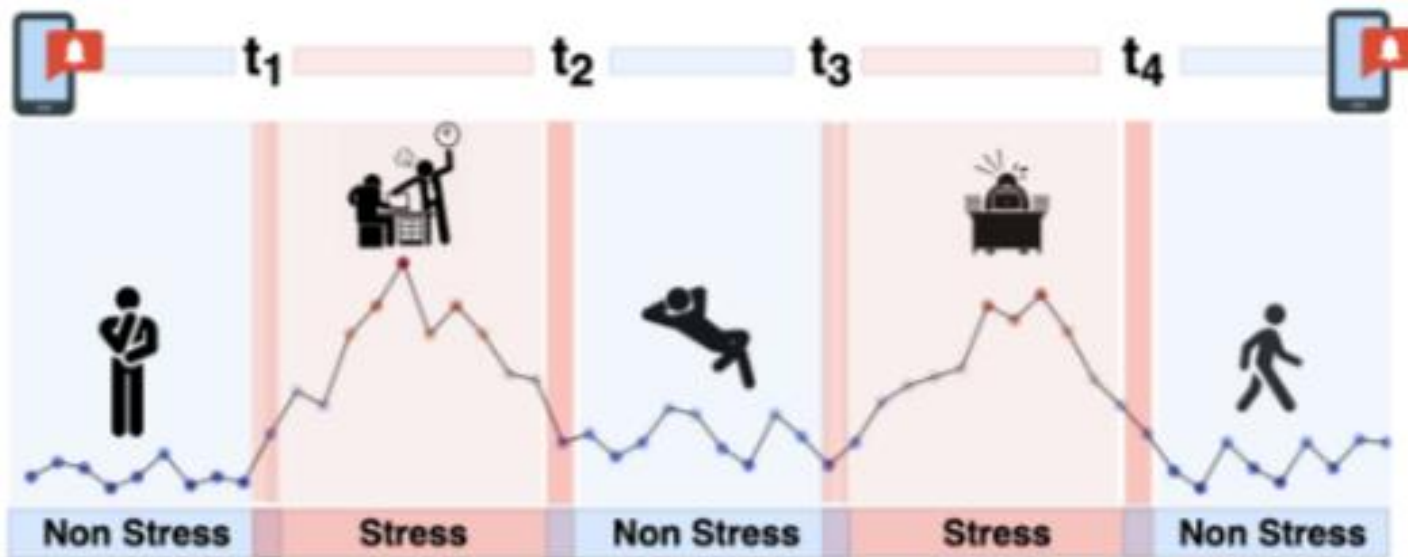
- ☐ Can system determine whether the response at t_γ refers?
 - ☐ Immediately preceding moment
 - ☐ Recent stress period
 - From t_α to t_β
 - ☐ Peak event itself
 - ☐ Some broader recollection of the recent past



* Causal ambiguity in EMA

◆ Label Smearing in Training Data

- When to ask becomes a first-order determinant
 - Supervision signal internally inconsistent
 - Associate one label with a mixture of incompatible patterns



* A single label collected at each trigger

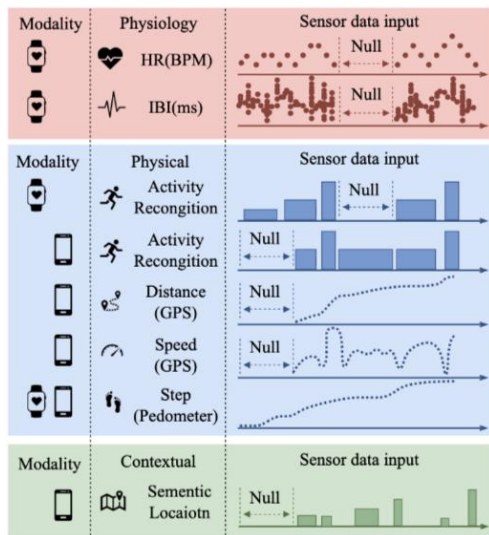
◆ Design Requirements for Event-Centric Self-Report

- ❑ **Three requirements for better sensor data labeling framework**
 - ❑ **Acquire self-reports closer** : physiologically meaningful events
 - ❑ **Reduce mixed-state labeling** : constructing training examples around candidate event intervals
 - ❑ **Reduce unnecessary interruption** : prioritizing candidate moments
- ❑ **Four stages in operating iEMA**
 - ❑ **Collects physiological, behavioral, and contextual signals** : constructs a temporally aligned multimodal sensing stream
 - ❑ **Identifies candidate informative moments** : comparing observed physiological responses
 - ❑ **Applies selective querying policy** : prioritizes only those with high expected information value
 - ❑ **Update the training set** : refine subsequent prompting and classification decisions

◆ Raw Heterogeneous Input Collection

□ Three broad categories of input

- **Physiological data** : heart rate (HR) and inter-beat interval (IBI)
- **Physical data** : activity recognition outputs, GPS-derived distance and speed, and step counts
- **Contextual data** : semantic location

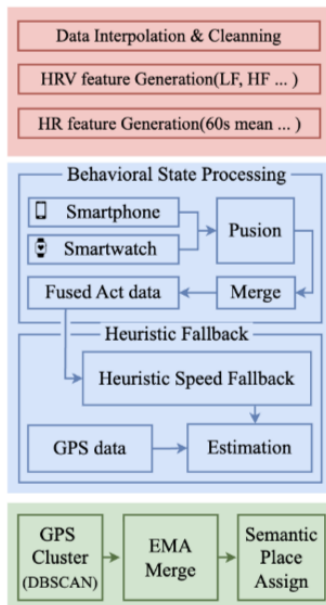


* Raw Heterogeneous Input Collection

◆ Feature Extraction from Physiological and Contextual Streams

□ Two complementary types of features

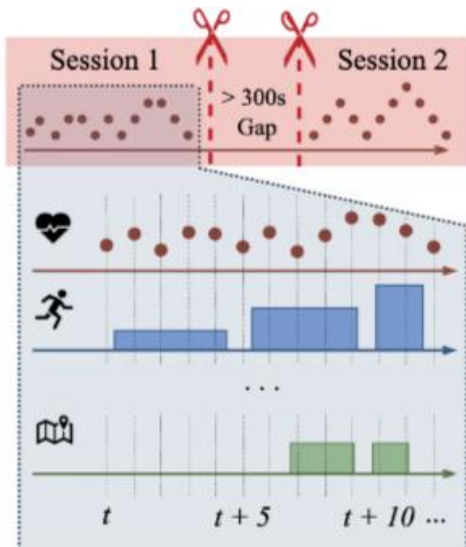
- **HRV features** : computed over a 60-second sliding window from IBI data when available
- **HR time-series features** : capture short-term trend and inertia



* Feature Extraction

◆ Sessionization and Shared Temporal Gridding

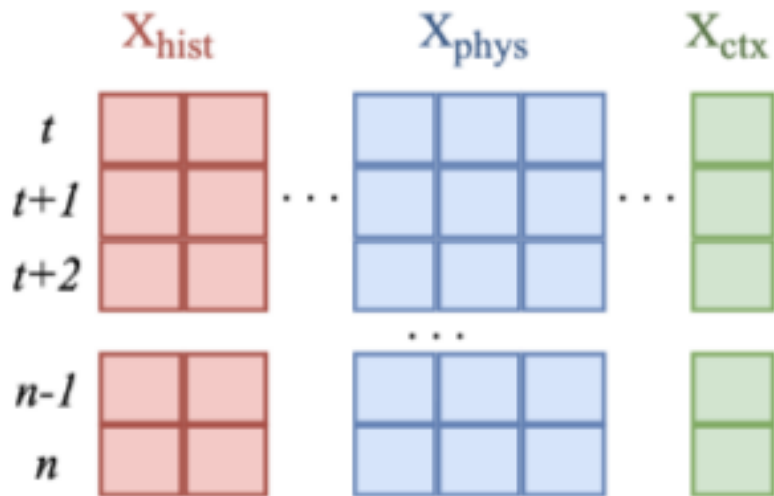
- ❑ iEMA organizes the raw timeline into valid sensing sessions
 - ❑ Creates a shared temporal grid at 1 Hz : resamples all asynchronous streams onto this common timeline
 - ❑ Physiological values, behavioral states, and contextual cues : interpreted relative to the same moment in time
 - ❑ Informative Moment detection : ask whether an observed physiological deviation is explainable



* Sessionization & Gridding

◆ Construction of the Aligned Multimodal Feature Matrix

- $V_t = [X_{hist}, X_{phys}, X_{ctx}]$
 - X_{hist} : captures short-term heart-rate trend through recent rolling statistics
 - X_{phys} : contains physical activity and mobility features
 - X_{ctx} : encodes contextual features such as semantic location



* Aligned Multimodal Features

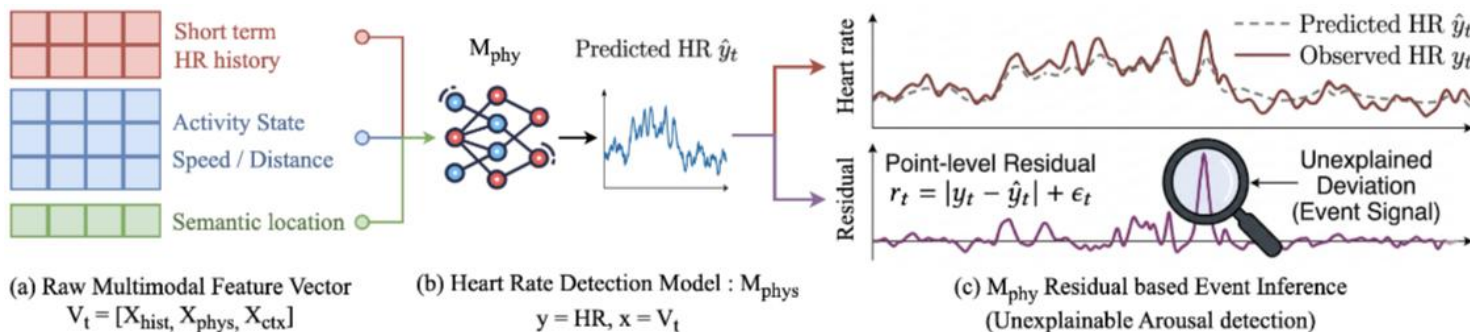
◆ Informative Moment Detection

□ Baseline Physical Model

- **Short-term HR history** : heart rate 5 seconds earlier and rolling statistics over the preceding 60 seconds
- **Activity State Speed / Distance** : contextual features
- **Semantic location**

□ Residual-Based Event Inference

- **y_t from M_{phys}** : computes a residual signal
- **Residual** : magnitude of unexplained physiological deviation
- **Residual magnitude** : event signal



◆ Informative Moment Detection

□ Window-level IM Formation

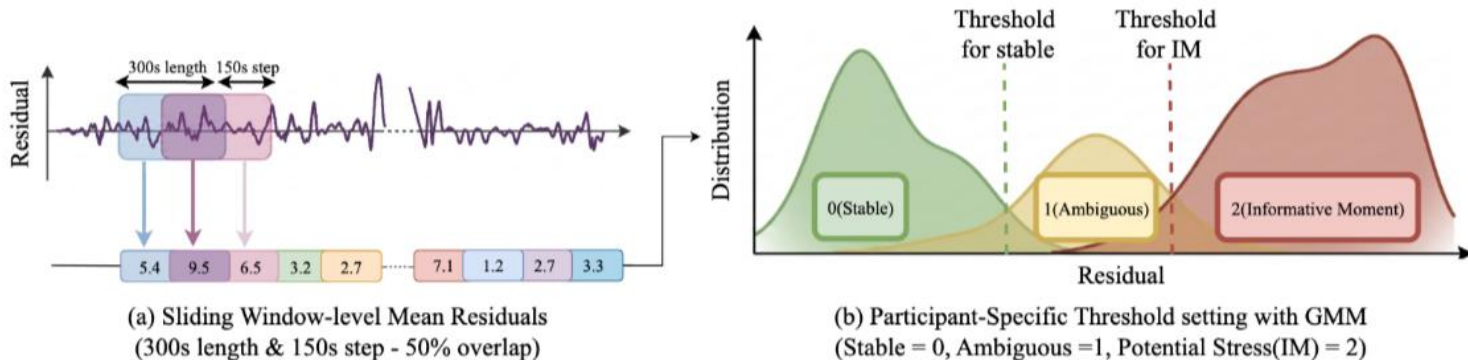
- Computes the mean residual : applies a sliding window of 300 seconds with a step size of 150 seconds

□ Personalized Thresholding with Gaussian Mixture Models(GMM)

- Three components to the distribution : window-level mean residuals

□ Point-level IM Assignment

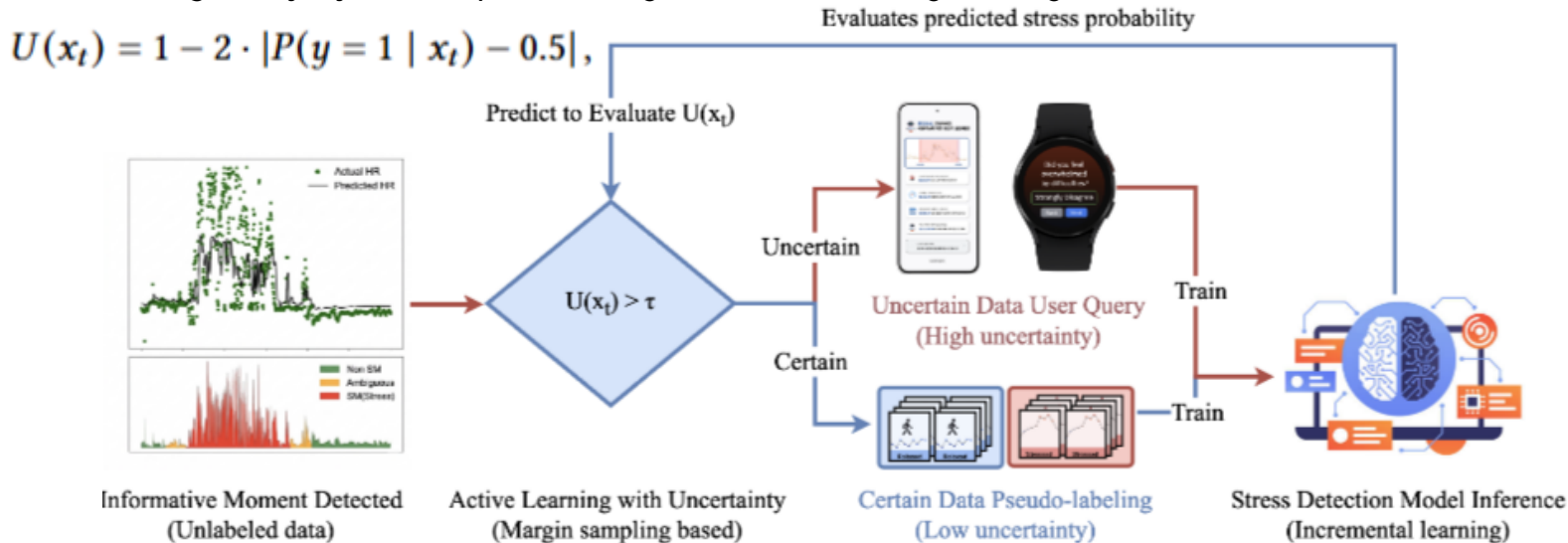
- 0 : stable
- 1 : ambiguous
- 2 : Informative Moment



* Personalized Thresholding with Gaussian Mixture Models(GMM)

◆ Active Incremental Experience Sampling(AIES)

- ❑ Event-level Hybrid Labeling : matches the practical unit of user interruption
- ❑ Uncertainty-aware Query Selection : uncertainty of the current downstream stress model
- ❑ Query Execution and Training-set Update : resulting label is applied to the full event segment
- ❑ Self-limiting Query Dynamics : preserves high-value labels during the stage

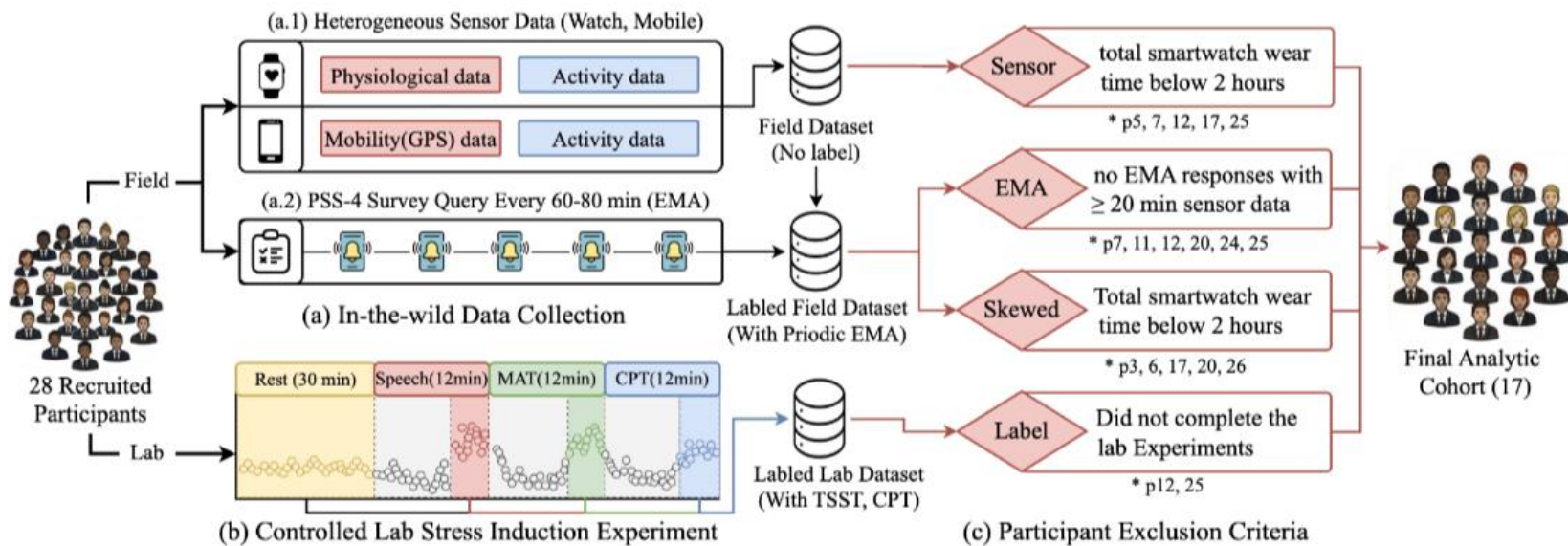


* EMA AIES(Active Incremental Experience Sampling)

◆ Dataset and Analytic Cohort

□ Whether IM-based label construction yields a more learnable training signal?

- **Heterogeneous Sensor Data** : Samsung Galaxy Watch 5 and an Android smartphone
- **PSS-4 Survey Query** : randomized intervals between 40 and 80 minutes



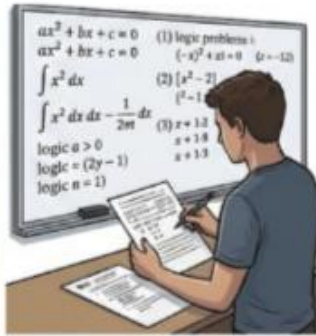
* Dataset construction and analytic cohort selection

◆ Dataset and Analytic Cohort

□ Why this separation is necessary?

□ Controlled Lab Stress Induction Experiment

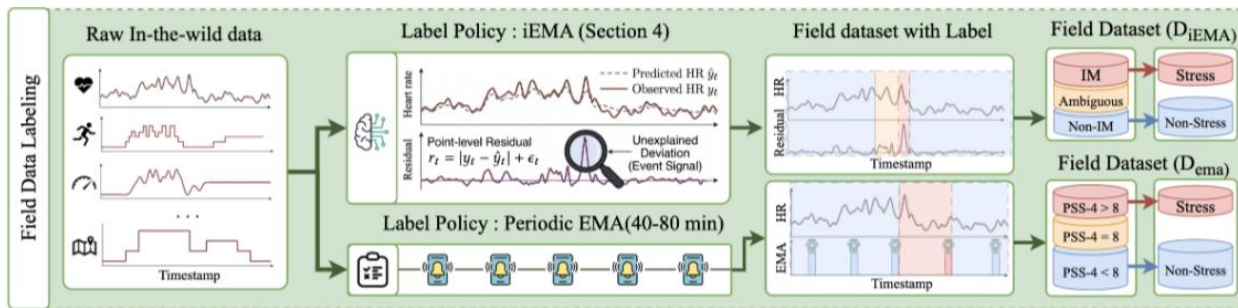
- Trier Social Stress Test (TSST) : public speaking and mental arithmetic tasks
- Cold pressor test (CPT) : controlled cold/pain stressor

TSST(Social stress)		CPT(Physical Stress)
Public Speaking Task	Mental Arithmetic Task	Cold Pressor Test
		
Social-evaluative stress through speech in front of evaluators	Cognitive stress through timed arithmetic under evaluation	Cold/pain stress through hand immersion in cold water

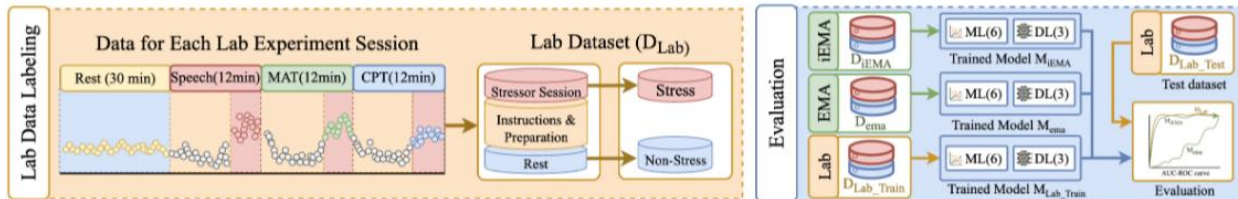
* Noisy in-the-wild sensing versus controlled laboratory stress reference

◆ Field-to-Lab Evaluation: Same-Timeline Label-Policy Comparison

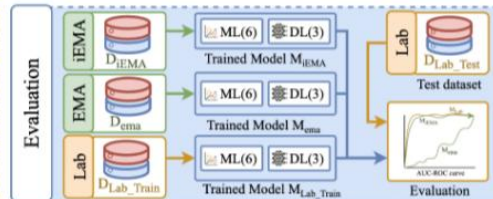
- Construct training label from this noisy daily-life substrate?
 - **Event-centric label construction**: Constructed from the same daily-life sensing timeline
 - **Informative Moments** : observed physiological response deviates strongly from this contextual baseline
 - **Downstream** : label-acquisition policy changes what can be learned from the same daily-life data



(a) Field dataset Labeling to D_{iEMA} , D_{ema}



(b) Lab Dataset (D_{Lab}) Labeling



(c) Downstream Stress Detection Model Evaluation

◆ Research Questions, Baselines, and Metrics

□ Three research questions that correspond to the main functional claims of iEMA

1) Do iEMA-derived event-centric labels produce more learnable field training signals than conventional periodic EMA labels when both are constructed over the same daily-life sensing timeline?

: use Area Under the ROC Curve (AUC-ROC) as the primary downstream stress-detection metric

2) Can context-aware residual modeling identify Informative Moments (IMs) that are distinguishable from ordinary physiological variation in daily-life sensing streams?

: report Mean Absolute Error (MAE) of the physiological baseline model across activity states and IM levels

3) Can Active Incremental Experience Sampling (AIES) reduce simulated query demand while preserving most of the downstream performance gained from event-centric label construction?

: report downstream AUC together with data usage.

◆ Field-to-Lab Learnability with Lab-to-Lab Reference

□ Three main patterns

□ **Lab-to-Lab** : average AUC-ROC of 0.847

□ **Field-to-Lab baseline** : average AUC-ROC of 0.501

□ **iEMA-derived field** : average AUC-ROC of 0.797

$$\frac{0.797 - 0.501}{0.847 - 0.501} \approx 0.856.$$

Evaluation	Training Label Source	Downstream Task: Stress Detection Model									Avg
		SVM	LR	RF	GBRT	XGB	ADA	LSTM	MLP	TRSF	
Lab-to-Lab	Lab-supervised	0.842	0.851	0.885	0.858	0.862	0.863	0.821	0.791	0.852	0.847
	Baseline (EMA)	0.458	0.566	0.501	0.541	0.520	0.534	0.449	0.541	0.399	0.501
Field-to-Lab	iEMA (SVR)	0.820	0.880	0.775	0.815	0.744	0.808	0.866	0.806	0.825	0.815
	iEMA (LINREG)	0.829	0.874	0.768	0.746	0.742	0.792	0.862	0.813	0.846	0.808
	iEMA (RF)	0.826	0.879	0.768	0.784	0.718	0.792	0.869	0.798	0.839	0.808
	iEMA (GBRT)	0.828	0.878	0.772	0.725	0.728	0.766	0.865	0.791	0.849	0.800
	iEMA (XGB)	0.822	0.879	0.764	0.725	0.724	0.762	0.863	0.795	0.843	0.797
	iEMA (ADA)	0.749	0.854	0.700	0.700	0.651	0.735	0.828	0.777	0.792	0.754
	iEMA (LSTM)	0.787	0.873	0.776	0.777	0.758	0.788	0.851	0.806	0.832	0.805
	iEMA (MLP)	0.807	0.879	0.747	0.714	0.697	0.786	0.866	0.785	0.853	0.793
	iEMA (TRSF)	0.773	0.868	0.773	0.735	0.745	0.766	0.853	0.774	0.832	0.791
	iEMA Avg.	0.805	0.874	0.760	0.747	0.723	0.777	0.858	0.794	0.835	0.797

◆ Informative Moment Validity

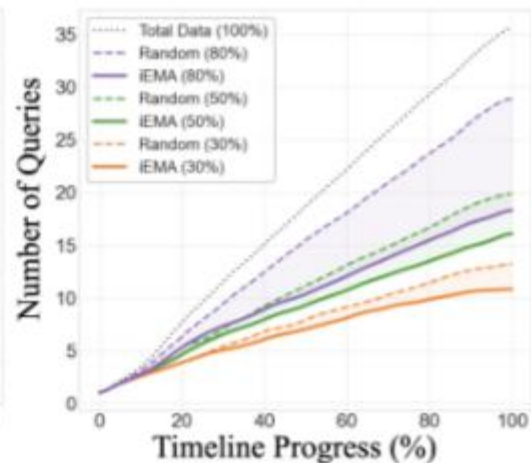
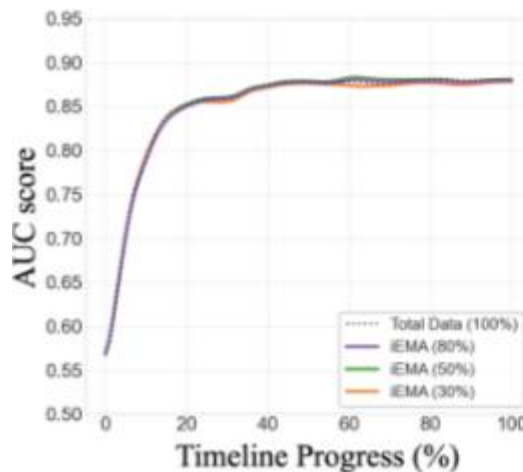
- ❑ ML/DL baseline models
 - ❑ non-IM periods : average MAE increases from 4.51 bpm
 - ❑ ambiguous periods : 9.86 bpm
 - ❑ IM periods : 19.30 bpm
- ❑ iEMA not equate high heart rate with stress
- ❑ Context-aware residual modeling can identify intervals where observed physiology departs

Type	Model	Activity States							IM Level			
		STILL	WALK	RUN	BIKE	VEH	UNK	Avg	Non-IM	Ambiguous	IM	Avg
ML	LinReg	7.82	10.09	12.38	12.04	9.01	8.19	9.92	3.63	8.30	16.07	9.33
	RF	7.76	10.04	12.71	11.27	9.05	5.02	9.31	3.48	8.56	17.06	9.70
	GBRT	7.79	10.04	12.64	11.83	9.10	5.80	9.53	3.48	8.58	17.06	9.71
	XGB	8.16	10.50	13.73	14.31	9.96	5.42	10.35	3.66	9.07	18.39	10.37
	SVR	7.71	10.12	14.02	15.13	9.89	37.76	15.77	3.40	8.62	17.65	9.89
	Ada	12.36	14.48	15.67	17.41	14.21	17.10	15.21	7.44	13.76	22.15	14.45
DL	LSTM	8.14	10.55	13.39	13.61	9.71	-	11.08	3.81	9.36	18.41	10.53
	TRSF	10.10	12.29	15.89	14.41	12.07	-	12.95	5.88	11.86	21.40	13.05
	MLP	8.83	12.83	15.43	(554.58)	13.08	(1262.36)	12.54	5.79	10.67	25.51	13.99
Average		8.74	11.22	13.98	13.75	10.68	13.21	11.85	4.51	9.86	19.30	11.22

◆ Simulated Query Reduction and Performance Retention

- AIES preserves performance close to the full-data reference
 - $\tau = 0.3$ setting : AUC-ROC of 0.8783 (using only 30.4% of the available data)
 - 99.8% of the full-data : AUC-ROC (0.8783/0.8801)

Strategy	Threshold	AUC	Data Usage (%)
Full Data	-	0.8801	100.0%
Random	($p = 0.3$)	0.8754	37.2%
AIES	($\tau = 0.3$)	0.8783	30.4%
Random	($p = 0.5$)	0.8805	55.8%
AIES	($\tau = 0.5$)	0.8803	45.2%
Random	($p = 0.8$)	0.8810	81.2%
AIES	($\tau = 0.8$)	0.8792	51.5%



◆ Why this finding matters for?

☐ Lab-to-Lab Reference Reveals a Label-Learnability Bottle

- ☐ **Field labels constructed by periodic EMA** : too temporally smeared or internally mixed
- ☐ **Event-centric labels** : better transfer to laboratory stress episodes across classifier families
- ☐ **Labels used to train it** : temporally aligned and learnable in the first place

☐ Label Timing and Alignment as a Systems Problem

- ☐ **Self-report timing** : treated as part of the sensing system
- ☐ **Aligning heterogeneous sensing streams** : using selective querying to decide which events are worth user input

☐ Broader Scope: Event-Centric Label Acquisition Beyond Stress

- ☐ **Residual-based IM module** : identifies moments under the current physical and contextual baseline

◆ Four limitations must be addressed before iEMA can be treated as a mobile mental-health sensing framework

☐ Prospective and Ecological Validation

- ☐ The current evaluation : same-timeline simulation
- ☐ Field-to-Lab evaluation : relies on laboratory stress episodes
- ☐ Prospective field validation : evaluate response latency, response rate, perceived burden, subjective relevance

☐ Semantic Ambiguity of Informative Moments

- ☐ Semantic interpretation of IMs
- ☐ Unexplained physiological arousal : reflect distress, excitement, anxiety, cognitive load, social pressure, fatigue

☐ Personalization, Cold Start, and Parameter Adaptation

- ☐ iEMA relies on personalized baseline modeling

☐ Safe and Context-Aware Prompt Delivery

- ☐ Event-centric prompting : not be equated with immediate interruption

◆ Event-centric framework for self-report acquisition in mobile stress sensing

- ❑ **Event-centric process** : should be aligned with physiologically meaningful moments
 - ❑ **Residual-based Informative Moment detection**: uncertainty-aware query selection
- ❑ **iEMA-derived event-centric labels** : improved average Field-to-Lab AUC-ROC to 0.797 (Lab-to-Lab 0.847)
 - ❑ **Context-aware residual modeling** : candidate IM intervals contained larger unexplained physiological deviations
 - ❑ **AIES retained 99.7% of the full-data AUC** : using 30.4% of event-centric samples (corresponding to a 69.6% reduction)